3D Structural and Petrophysical Characterization of the Hugin Reservoir

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Volve field Platform (Photo: Colin Dobinson, Source: https://www.equinor.com/news/archive/2008/03/06/Volveviservei)

1. Executive Summary

This study presents a well-based structural and petrophysical characterization of the Hugin reservoir using data from the Equinor Volve Data Village. The analysis focuses on well 15/9-19 BT2 and integrates Hugin top and base interpretations, well-survey data, official fault interpretations, and interpreted petrophysical logs to establish the structural position and reservoir-property variability along the well trajectory.

Quality control and depth-reference reconciliation were performed before integrating the datasets. The BT2 well survey was used to derive TVDSS and spatial coordinates along the measured-depth trajectory, while the Hugin formation interval was constrained using interpreted top and base picks. The survey-derived depths reproduce the independently reported Hugin picks to within approximately 0.1 m vertically, providing a strong internal check on the depth integration.

Within the interpreted Hugin interval, effective porosity (PHIF) is centered around 20%, with a median of 0.206 V/V and a relatively narrow interquartile range. Interpreted shale volume (VSH) is generally low to moderate but more variable, while horizontal permeability (KLOGH) exhibits substantially greater heterogeneity, with values spanning several orders of magnitude. Water saturation (SW) is predominantly high, with a median of 1.0 in the interpreted petrophysical model. Crossplots of PHIF–VSH, PHIF–KLOGH, and PHIF–SW were used to investigate relationships between the interpreted properties and identify trends within the Hugin interval.

The resulting 3D structural visualization places the BT2 trajectory within the interpreted Hugin framework, while a regional 2D fault map provides structural context for nearby interpreted fault features. Because the detailed petrophysical analysis is based on a single well and the available fault interpretations are primarily in the time domain, the study does not attempt to construct a reservoir-wide 3D property model or make definitive compartmentalization claims. Instead, it establishes a reproducible well-based characterization workflow and identifies the additional structural, well, and seismic information required for a more complete reservoir-scale model.

2. Dataset & Context

Dataset context:

Equinor describes Volve as a decommissioned field in the central North Sea, approximately 5 km north of Sleipner Øst in block 15/9, with about 80 m water depth. Production was from Jurassic Hugin Formation sandstone at approximately 2750–3120 m TVDSS. The Volve Data Village release contains field-related production, well, completion, seismic, well-log, geological/stratigraphic and modelling data. The present study uses only a small subset relevant to the BT2 Hugin characterization.

Dataset Source

The data were accessed through the Databricks dataset page on September 28, 2026

Databricks Catalog volume: /Volumes/equinor_asa_volve_data_village/public/volvezipfiles

Given volume of data, only files of interest are picked up through Databricks python notebooks.

Picked up files directory structure of Databricks Catalog is reproduced locally

Files are inventoried and listed in data_inventory.csv

The original dataset is available through Databricks Marketplace

The study integrates selected structural and well-log data from the Volve Data Village to characterize the Hugin interval intersected by 15/9-19 BT2. The work is intentionally a well-based characterization rather than a field-scale reservoir model.

1. The Hugin reservoir is a Jurassic sandstone reservoir in the Volve field in the central North Sea. This study characterizes the structural position and selected petrophysical properties of the Hugin interval sampled by well 15/9-19 BT2. The objective is to integrate Hugin top/base horizon interpretations, the BT2 well trajectory and interpreted petrophysical logs into a consistent 3D well-based characterization, while avoiding unsupported reservoir-wide property extrapolation.

2. Well 15/9-19 BT2 was selected because it has explicit Hugin top and base picks in the Volve well-picks dataset and a matching standard well survey and CPI petrophysical interpretation. The availability of MD, TVD/TVDSS and horizontal-position information allows the well trajectory to be tied independently to the published Hugin picks and then used as the spatial reference for the petrophysical visualization.

3. The scope is deliberately limited to structural context and petrophysical characterization along a single well. The study does not construct a reservoir-wide 3D property grid, estimate lateral property distributions, or claim fault compartmentalization from the available data. The principal outputs are the Hugin top/base structural surfaces, the BT2 trajectory and well-pick tie, a regional 2D fault context, well-log characterization over the picked interval, and 3D visualization of interpreted properties along the BT2 trajectory.

3. Data and Quality Control

1. The principal inputs are the Hugin top and base horizon files, Volve well picks, the 15/9-19 BT2 standard well survey, official 2011 fault sticks, and the 15/9-19 BT2 CPI LAS file. These were selected because they provide complementary structural, trajectory and interpreted petrophysical information for the selected well.

2. Coordinate system, units and depth references:

Distances and coordinates are handled in metres. Survey measured depth and vertical depth are reported relative to the rotary kelly bushing (RKB), while the horizon and well-pick depth reference is treated as TVDSS after reconciliation of sign convention. The survey metadata specify KB-MSL = 25 m.

Depth references:

Petrophysical interpretations are recorded in measured depth (MD).
The BT2 survey provides MD-RKB, TVD-RKB, and absolute UTM Easting/Northing
The Hugin horizon files use an SSTVD-below-sea-level positive-down convention. For comparison with the well survey and published well picks, unsigned TVDSS is used, where depths below MSL are positive.

For the BT2 survey, TVDSS = TVD-RKB − 25.0 m because the survey metadata specify RKB = 25 m above MSL.

LAS MD is matched directly to the survey MD-RKB coordinate, and TVD-RKB is obtained by linear interpolation of the actual survey stations. TVDSS is then calculated as TVD-RKB − 25.0 m.

Hugin top/base TVDSS picks closely:

At the Hugin top and base MD picks, the survey-derived TVDSS reproduces the published well-pick TVDSS closely. For the base, the interpolated signed TVDSS is −3274.58877 m versus −3274.68 m in the well-picks file (absolute residual 0.09123 m). For the top, it is −3148.83476 m versus −3148.92 m (absolute residual 0.08524 m). This provides an independent check of the depth-reference conversion.

well identifier	formation	picks MD	picks TVDSS	survey derived TVD RKB	TVDSS interpolated	TVDSS residual
15/9-19 BT2	Hugin Base	4178.59	**-3274.68**	3299.58877	**-3274.58877**	**0.09123**
15/9-19 BT2	Hugin Top	4035.80	**-3148.92**	3173.83476	**-3148.83476**	**0.08524**

3. Petrophysical curve inventory and quality control - missing values, valid ranges and flagged anomalies:

Curves Depth Range: [3768.09,4249.8264]
NULL -999.25 NULL VALUE

QC note: VSH is reported with a maximum of 17.1151 V/V, which is outside the conventional 0–1 range for a volume fraction. This is treated as an unresolved curve-definition or data-quality anomaly and is not silently clipped.

15/9-19_BT2 well log data summary

Mnemonic	Unit	Valid sample count	Missing Count	Minimum	Maximum
DEPTH	M	3162	0	3768.09	4249.8264
BVW	V/V	2926	236	0	0.250109
CARB_FLAG	UNITLESS	3015	147	0	1
COAL_FLAG	UNITLESS	3162	0	0	1
KLOGH	MD	2926	236	0	13705.3096
KLOGV	MD	2921	241	0	13974.7588
PHIF	V/V	2926	236	0	0.295406
RHOFL	G/CM3	3162	0	0.9	0.9
RHOMA	G/CM3	2926	236	2.64	2.71
RW	OHMM	2926	236	0.019953	0.0214721
SAND_FLAG	UNITLESS	2926	236	0	1
SW	V/V	2926	236	0.117	1
TEMP	DEGC	2926	236	113.7916	124.0921
VSH	V/V	2926	236	0.0199	17.1151

4. No material unresolved MD-to-TVDSS uncertainty was identified for the BT2 well. The main remaining convention to document is the sign difference between the horizon-file positive-down SSTVD values and signed TVDSS values used for the well. The official fault sticks remain in two-way time (TWT), so their vertical relationship to the depth-converted Hugin surfaces is unresolved.

4. Methodology

1. Horizon and fault data preparation and plotting:

   Horizons files:

   - Consist of 2 dat files for hugin top and base formation:
     Hugin_Fm_Base+ST10010ZC11_Near_190314_adj_2999_EasyDC+STAT+DEPTH.dat
     Hugin_Fm_Top+ST10010ZC11_Near_190314_adj2_2760_EasyDC+STAT+DEPTH.dat
   - Data extraction is done via rigid regex pattern matching:

The relevant horizon records were verified against the file headers as line, trace, X, Y and Z values; the numerical records were then extracted for analysis.

A strict regular-expression pattern was used for the comma-separated numerical horizon records, followed by conversion to floating-point values.

- A dictionary of lines is created and converted to pandas data frame for further analysis.
- Numerical columns are converted to float.
- Horizon 2D and 3D surfaces are plotted from the extracted X, Y and Z coordinates using scatter plot. Depth colorbars are normalized (using Normalize() and ScalarMappable) to the corresponding horizon range; triangulated surfaces are used where a continuous 3D visualization is required.
-
- Triangulated surfaces are used for continuous 3D visualization; the triangulation is a visualization of the interpreted samples rather than a new geological property model.

Faults files:

File checked: Volve_Official_Faults.dat

- No pattern matching is used as it only consists of data lines starting from the first line through the end.
- Fault records were parsed into observation number, fault name and year, using str.extract(pattern) : Obs-Name-Date/Year. Only the 2011 interpretation was selected for the structural-context map.
- Lines are stored in a dictionary and then converted to dataframe
- Numerical columns are converted to float.
- Distinct colors are used to separate fault-stick groups visually; the color assignment using colorsys is a plotting aid and has no geological meaning.
- Fault sticks are grouped by fault name and observation/stick identifiers and plotted as separate line segments. Because their vertical coordinate is TWT rather than depth, they are used for 2D horizontal structural context and proximity screening, not as depth-converted 3D fault surfaces.
-

2. Well survey and Petrophysical interpretations data preparation and plotting:

   Well survey file:

   File inspected: 15_9-19 BT2_Standard_Survey_NPD.txt

   - A strict pattern is used to extract data lines of interest:

   "^\d+\.\d+\s+\d+\.\d+\s+\d+\.\d+\s+\d+\.\d+\s+-?\d+\.\d+\s+-?\d+\.\d+\s+\d+\.\d+\s+\d+\.\d+\s+\d+\.\d+\s*$"

   MD-RKB, TVD-RKB, and absolute UTM Easting/Northing coordinates are extracted and stored in a dataframe. The survey metadata identify the coordinate system as UTM Zone 31N and specify KB-MSL = 25 m.

   - 3D and 2D Well trajectory line are plotted
   - A TVDSS column is added using TVDSS = TVD-RKB − 25 m. The survey MD-RKB coordinate is retained unchanged for matching the LAS MD; no 25 m correction is applied to MD.

   Petrophysical Interpretations files:

   - File checked is las file: 15_9-19_BT2_CPI.las
   - The LAS headers are inspected and the well information, curve mnemonics, units, descriptions and depth coverage are recorded in the project data dictionary as follows:

   Well information
   For every curve: mnemonics, units and description
   Curves depth range

   - Data is directly extracted and converted to a dataframe using lasio and las.df()
   - The LAS null value −999.25 is replaced with NaN before numerical analysis and plotting. Curve-level valid counts, missing counts and ranges are retained for quality control.
   - Subplots is used to plot all petrophysical curves

3. Survey interpolation and comparison with Hugin picks

Well_picks_Volve_v1.dat is checked for well picks.
Data is gathered for selected 15/9-19 BT2 as following:

well identifier	formation	observation number	MD	TVD	TVDSS	Easting	Northing
NO 15/9-19 BT2	Hugin Fm. VOLVE Base	1	4178.59	3299.68	-3274.68	436437.4	6477478.2
NO 15/9-19 BT2	Hugin Fm. VOLVE Top	1	4035.8	3173.92	-3148.92	436408.4	6477539.1

The BT2 Hugin top and base picks are compared with the survey by interpolating the survey's TVDSS and X/Y coordinates at the pick TVDSS. Residuals are calculated against the published pick values.

Interpolated hugin top and base well picks (X,Y) coordinates by well trajectory:

15/9-19 BT2 Hugin Base Pick Interpolated Well pick (X,Y):
[436439.01438, 6477478.36630]
(X,Y) Residual: [1.61438176 0.16630393]

15/9-19 BT2 Hugin Top Pick Interpolated Well pick (X,Y):
[436410.06785, 6477539.28719]
(X,Y) Residual: [1.66785377 0.18719345]

The interpolated trajectory coordinates are close to the published Hugin pick coordinates, with horizontal residuals of about 1.6–1.7 m in Easting and less than 0.2 m in Northing. Together with the independent TVDSS check, this supports using BT2 as the trajectory for the well-based characterization.

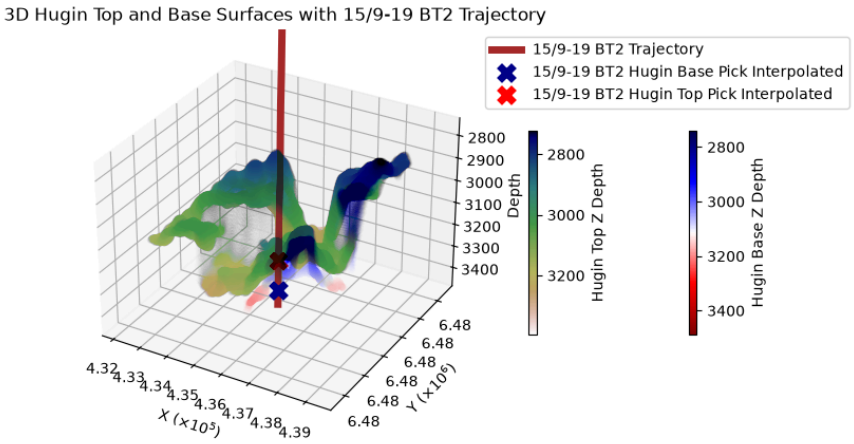


Figure 4.1. 3D Hugin Top and Base Surfaces - 15-9-19 BT2 Well picks interpolated

4. Petrophysical curve selection, interval filtering, crossplots and summary statistics.
   The petrophysical dataframe is screened for valid samples and the DEPTH column is retained as the LAS MD reference. Curves are looped for overview plotting, after which the Hugin top/base MD interval is used for focused characterization.
   The full LAS range is first inspected to identify data coverage and missing-value patterns. The detailed reservoir interpretation is then focused on the MD interval between the published Hugin top and base picks.

5. 3D visualization workflow.
   3D structural mapping:
   The Hugin top and base surfaces are plotted first, followed by the published Hugin picks and their survey-interpolated trajectory positions. The full BT2 trajectory is then added, with the Hugin interval identifiable within it.

   3D petrophysical characterization:
   The interpreted curves are mapped onto the BT2 survey trajectory by interpolating each petrophysical curve from LAS MD onto survey MD. The survey supplies TVD-RKB, TVDSS and X/Y at those trajectory points; therefore the petrophysical values are not independently extrapolated in TVD.
   All available interpreted curves are processed for visualization. Emphasis is placed on the reservoir-relevant properties and indicators.

Trajectory points within the petrophysical coverage are represented as stacked line segments using the survey X, Y and vertical coordinates.

For the final 3D property trajectory, LAS MD is interpolated onto survey MD; survey MD is then associated with the measured TVD-RKB and X/Y coordinates, and TVDSS is computed from the 25 m RKB-MSL datum.
For each property, valid LAS samples are interpolated in MD onto the survey trajectory. The resulting property values are assigned to a Line3DCollection so that the BT2 trajectory can be colored continuously by the selected property. Values outside the LAS coverage are not extrapolated.
A colorbar is added for continuous petrophysical variables; categorical flags are better represented with discrete labels or legends rather than continuous color scales.

5. Results

1. BT2 trajectory and Hugin top/base tie:

   Interpolated well picks depth by Hugin horizons:

   The Hugin surface-interpolated base position at MD 4178.59 m gives TVDSS ≈ 3273.01468 m. The published signed base pick TVDSS is −3274.68 m, giving an absolute residual of ≈ 1.665 m.
   TVD Residual difference: 26.665311036656476,
   TVDSS Residual difference: 1.665311036656476

   The Hugin surface-interpolated base position at MD 4035.80 m gives TVDSS ≈ 3147.52239 m. The published signed base pick TVDSS is −3148.92 m, giving an absolute residual of ≈ 1.397 m.

   TVD Residual difference:26.397604917399804,
   TVDSS Residual difference:1.397

Using the consistent signed TVDSS convention, both Hugin picks are reproduced by the hugin horizons to better than 1.5 m vertically .

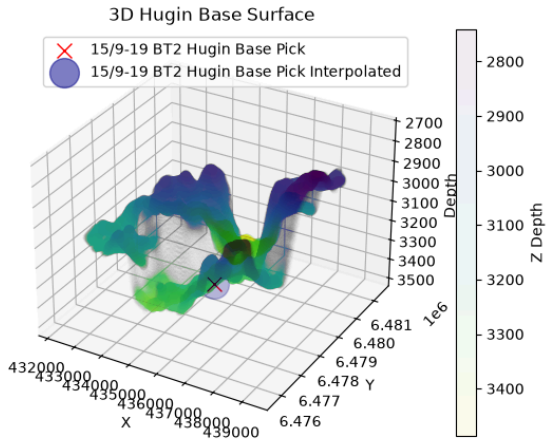


Figure 5.1. Hugin Base Interpolation of 15/9-19 BT2 Hugin Base Pick

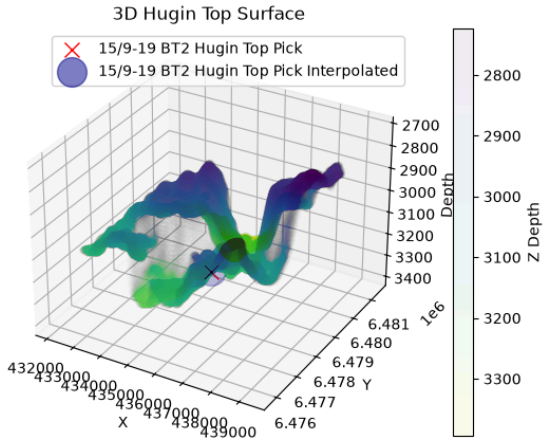


Figure 5.2. Hugin Top Interpolation of 15/9-19 BT2 Hugin Top Pick

The agreement of both Hugin picks with the independently interpolated survey TVDSS, together with the interpolated Hugin horizons and with the small horizontal coordinate residuals, provides a strong internal consistency check for selecting 15/9-19 BT2 for this well-based study.

2. Structural context and nearby interpreted faults:

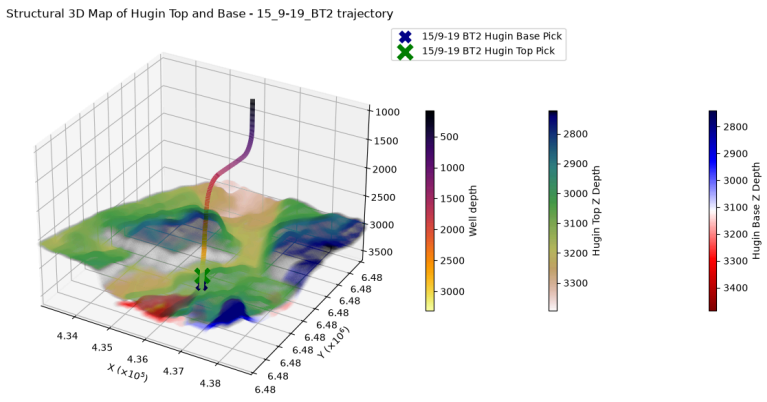


Figure 5.3. Hugin Top and Base Surfaces and 15/9-19 BT2 well trajectory

The regional 2D structural map shows the selected 2011 fault-stick interpretation in the vicinity of the BT2 trajectory and Hugin picks.

Horizontally nearby fault-stick groups include:

1-Volve-F11, 2-Volve-F11, 3-Volve-F11, 1-Volve-F29, 3-Volve-F29, 1-Volve-F31, 2-Volve-F31, 2-Volve-F35 and 3-Volve-F35.

Fault-stick groups that are horizontally close to the BT2 trajectory and Hugin pick locations include:

2-Volve-F11 and 3-Volve-F11 are among the closest candidates in the 2D screening.

These are proximity-screening results in the horizontal coordinate system only. Because the fault-stick vertical coordinate is TWT and no validated TWT-to-depth relationship was established, the study does not claim that these faults intersect the BT2 well or Hugin interval in 3D.

3. Within the Hugin picked interval, the interpreted curves show vertical variation in shale volume, water saturation, porosity, bulk-volume water, density, temperature, resistivity and permeability. The following observations are descriptive of BT2 and should not be generalized laterally.

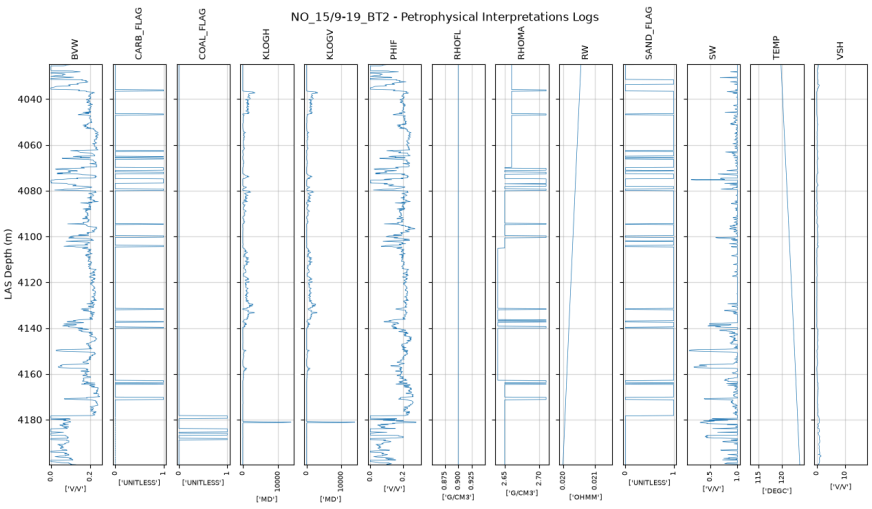


Figure 5.4. 15-9-19 BT2 Well log Petrophysical Interpretations

4. Crossplot patterns and interval statistics:

Hugin interval petrophysical statistics, 15/9-19 BT2

	PHIF	VSH	KLOGH	SW
count	937	937	937	937
mean	0.1981700726	0.1898261473	488.9278509	0.9509041622
std	0.0429973362	0.1375961045	622.9849236	0.1250739584
min	0	0.0199	0.0001	0.117
25%	0.189585	0.0763	3.9479	0.9577
50%	0.206063	0.1476	199.1382	1
75%	0.22369	0.3023	822.4001	1
max	0.272434	0.6922	3445.3123	1

The interpreted petrophysical properties within the Hugin interval show moderate porosity, variable shale volume, strongly heterogeneous permeability, and predominantly high interpreted water saturation. PHIF ranges from 0 to 0.272 V/V, with a mean of 0.198 V/V and median of 0.206 V/V. The relatively narrow interquartile range of 0.190–0.224 V/V indicates that porosity is concentrated around approximately 20% through much of the interval.

VSH exhibits greater variability, ranging from 0.020 to 0.692 V/V, with a median of 0.148 V/V and an interquartile range of 0.076–0.302 V/V. This indicates predominantly low-to-moderate interpreted shale volume, but with more heterogeneous shale content than porosity.

KLOGH shows the greatest variability of the four properties. Its mean of 488.9 mD is substantially higher than the median of 199.1 mD, while the interquartile range extends from

3.95 to 822.4 mD and the maximum reaches 3445.3 mD. This strong spread and right-skewed distribution indicate pronounced permeability heterogeneity within the interpreted interval.

SW is strongly concentrated toward high values, with a mean of 0.951 and median of 1.0. More than half of the samples therefore have an interpreted water saturation of 1.0, although lower values occur locally, with a minimum of 0.117. These values describe the output of the interpreted petrophysical curves and should not be treated as an independent fluid-contact determination.

The KLOGH–PHIF crossplot indicates a positive association between porosity and interpreted horizontal permeability, although permeability displays considerably greater scatter. Samples with similar porosity can exhibit substantially different permeability, indicating that porosity alone does not fully explain the permeability variability.

Figure 5.3. 3D Hugin Top and Base Surfaces - 15-9-19 BT2 Well picks interpolated

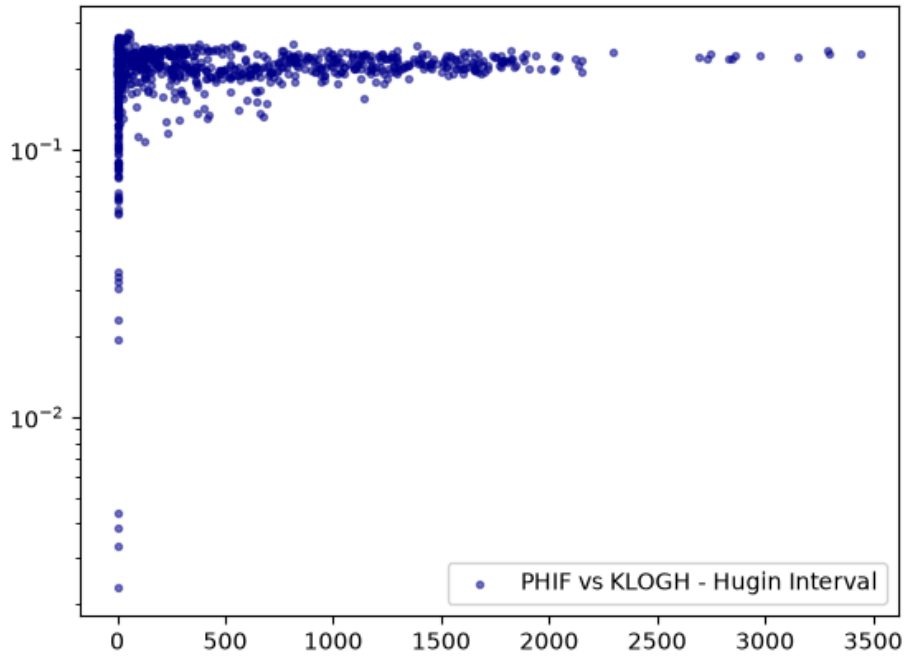


Figure 5.5. PHIF–KLOGH relationship for the Hugin interval in well 15/9-19 BT2 (logarithmic permeability scale)

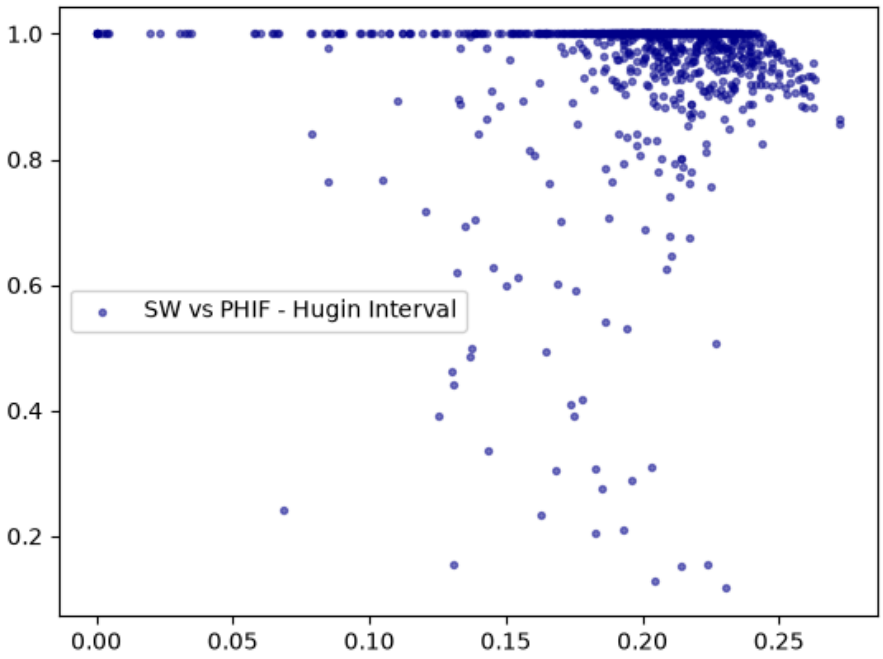


Figure 5.6. PHIF–SW crossplot for the Hugin interval in well 15/9-19 BT2

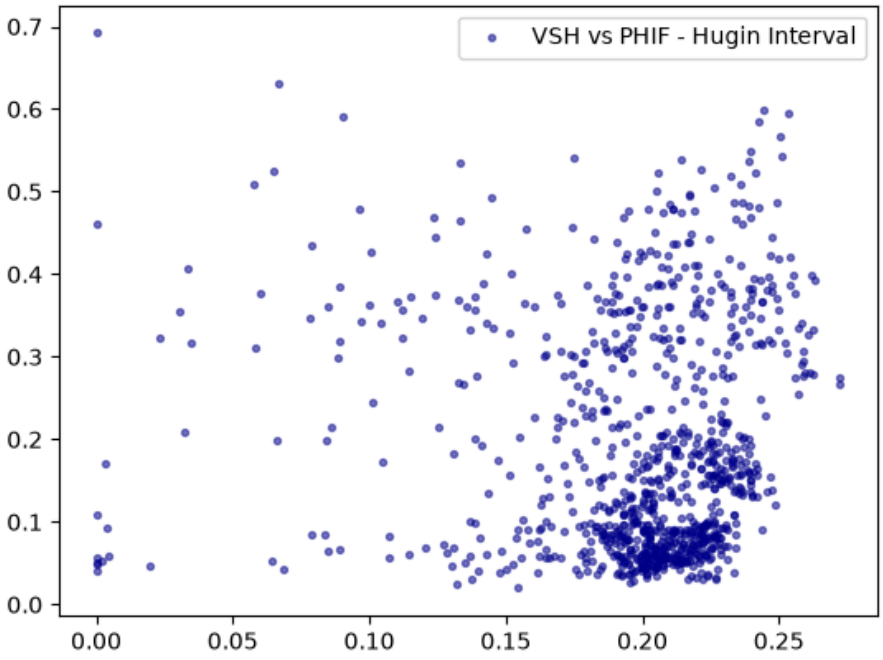


Figure 5.7. PHIF–VSH crossplot for the Hugin interval in well 15/9-19 BT2

5. Petrophysical Characterization

   The full LAS curve range is approximately MD 3768.09–4249.8264 m. The Hugin-specific interpretation should be restricted to the published top/base MD interval, approximately 4035.80–4178.59 m, while the full-range plot is retained for QC and context.

   When a property is interpolated onto the survey trajectory, the interpolation domain is the valid LAS coverage for that property. For example, the VSH data extend only to approximately MD 4213.86 m in the cleaned curve used here; values outside that domain should be left as NaN rather than extrapolated.

   Petrophysical interpretation curves in MD interpolated at well trajectory in MD:

MD	TVD-RKB	TVDSS	x	y	VSH
3783	2946.84	2921.84	436366.773	6477641.746	1.75227874
3814	2974.44	2949.44	436371.981	6477628.625	1.59241706
3843	3000.3	2975.3	436376.884	6477616.46	1.051466142
3872	3026.23	3001.23	436381.84	6477604.463	1.338754331
3900	3051.35	3026.35	436386.629	6477593.071	1.145168504
3929	3077.42	3052.42	436391.605	6477581.377	1.059524934
3957	3102.62	3077.62	436396.481	6477570.19	0.9754314961
3987	3129.67	3104.67	436401.71	6477558.334	0.9260173228
4015	3155	3130	436406.5	6477547.397	0.2488687664
4044	3181.26	3156.26	436411.452	6477536.141	0.04871023622
4073	3207.37	3182.37	436416.666	6477524.652	0.1024120735
4102	3233.15	3208.15	436422.331	6477512.648	0.3462958005
4140	3266.37	3241.37	436430.359	6477496.043	0.1010590551
4170	3292.23	3267.23	436437.01	6477482.377	0.3300496063
4197	3315.36	3290.36	436443.233	6477469.925	0.9600047244

3D Hugin Reservoir Petrophysical Characterization along 15_9-19_BT2 trajectory

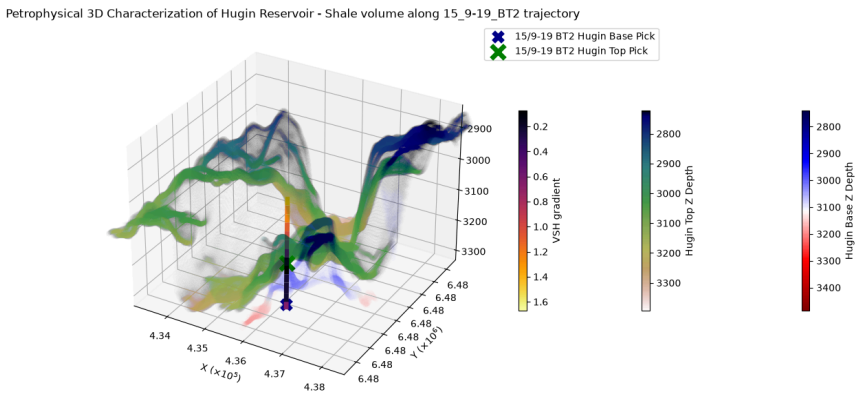


Figure 5.8. VSH distribution along the 15/9-19 BT2 trajectory within the Hugin interval

The BT2 trajectory shows measurable vertical variation in the interpreted properties across the Hugin interval. The observations below describe the sampled well response and are not a reservoir-wide property estimate.

- VSH generally decreases through the Hugin interval in the displayed trajectory, with a marked increase toward/near the upper part of the interval. However, the reported VSH maximum of 17.1151 V/V is physically inconsistent with a conventional volume-fraction interpretation and must be treated as a QC anomaly until its source/curve definition is verified. It should not be clipped to 1.0 without documenting the reason.
- BVW increases over portions of the picked interval, indicating vertical variation in the interpreted bulk-volume-water response.
- SW is generally high, often close to 1 V/V, with several sharp decreases. These local features can be visually muted when values are sampled only at relatively sparse survey stations, so the underlying LAS log should remain the reference for fine-scale variation.
- PHIF varies through the interval, with the available trajectory samples suggesting an average increase from roughly 0.13 toward 0.17 V/V in parts of the interval; this should be reported as a local well observation rather than an interval-wide average unless calculated from all valid LAS samples.
- RHOFL is constant at 0.9 g/cm³ in the provided interpretation and therefore provides little visual variation.
- RHOMA is approximately 2.65–2.66 g/cm³ through much of the displayed interval, with short-lived deviations.
- RW decreases approximately linearly while TEMP increases approximately linearly over the displayed trajectory segment.

Lithology Indicators:

- SAND_FLAG is predominantly 0 in the displayed samples, with local changes around the Hugin interval. Because the flag is categorical, it should be interpreted together with the other lithology indicators rather than as a continuous volume fraction.
- COAL_FLAG is predominantly 0, with isolated peaks near/below the Hugin base pick.
- CARB_FLAG contains isolated peaks; these are sparse and should be discussed as interpreted indicators rather than definitive lithological thicknesses.

Permeability:

- KLOGH and KLOGV show strong local variation, including increases and a sharp high-value feature near/below the Hugin base. Permeability spans several orders of magnitude in the LAS summary, so log-scale visualization is preferable and individual spikes should be checked against the original curve.

6. Discussion and limitations

The BT2 logs indicate a vertically heterogeneous interpreted interval. The response includes generally high water saturation, variable effective porosity and shale-volume indicators, and strongly variable permeability. The lithology flags are mostly inactive in the displayed trajectory samples, with isolated excursions. These observations describe the petrophysical interpretation at BT2 and do not establish net-to-gross, hydrocarbon saturation, or reservoir quality across the field.

With one well, the study can demonstrate the structural tie at BT2, describe vertical petrophysical variation, and show how interpreted properties can be positioned along a real 3D trajectory. It cannot reliably infer lateral continuity, field-wide property trends, compartmentalization, volumetric distributions, or a reservoir-wide 3D property model.

The Hugin horizon files provide spatial/depth samples but do not by themselves provide a validated TWT/depth conversion. The study therefore keeps the official fault sticks in their native TWT domain and uses them only for 2D structural context and horizontal proximity screening. A 3D fault intersection analysis would require a validated time-depth relationship or depth-converted fault interpretation.

The horizon vertical reference is treated consistently with the well picks after independent checking against the BT2 survey. The close top/base residuals reduce, but do not eliminate, uncertainty associated with file conventions and sign handling.

The CPI curves are interpreted products rather than raw measurements, and independent core or laboratory validation is not available in the present workflow. Curve-specific anomalies also remain: notably VSH values above 1 V/V require verification before VSH is used quantitatively. Petrophysical conclusions should therefore be treated as well-log interpretation results rather than independently validated reservoir properties.

7. Conclusions and next Steps

This study developed a well-based structural and petrophysical characterization of the Hugin reservoir using well 15/9-19 BT2 from the Equinor Volve Data Village. Hugin top and base interpretations, well-survey data, official fault interpretations, and interpreted petrophysical logs were integrated into a consistent structural framework and 3D visualization.

The depth-reference workflow provided a reliable integration between the BT2 well survey, petrophysical logs, and Hugin formation picks. Survey-derived TVDSS values reproduce the independently reported Hugin top and base picks to within approximately 0.1 m, supporting the consistency of the structural integration.

Petrophysical statistics indicate that the interpreted Hugin interval has moderate and relatively consistent porosity, with PHIF centered around 20%, while VSH indicates generally low-to-moderate but variable shale content. In contrast, KLOGH exhibits substantially greater variability, ranging from very low values to several thousand mD. This indicates that permeability heterogeneity is considerably stronger than the observed porosity variability within the investigated interval. SW is predominantly high, with a median interpreted value of 1.0, although locally lower values occur.

The PHIF–VSH crossplot was used to examine the relationship between interpreted porosity and shale volume, providing evidence for the degree of lithological and petrophysical heterogeneity within the interval. The PHIF–KLOGH crossplot demonstrates the broad permeability response associated with the observed porosity range and highlights that similar porosity values can correspond to substantially different permeability values. This emphasizes that porosity alone is insufficient to describe the flow-property variability of the investigated interval. The PHIF–SW crossplot further illustrates the relationship between pore volume and the interpreted saturation response, with most observations concentrated toward high water saturation.

The combined structural and petrophysical results provide a consistent well-scale characterization of the Hugin interval in BT2, rather than a reservoir-wide property model. The regional fault interpretation provides structural context, but the available fault data are primarily in the time domain and were therefore not used to make definitive three-dimensional fault-intersection or compartmentalization claims.

A reservoir-scale characterization would require additional wells, validated depth-converted fault and seismic interpretations, and sufficient spatial coverage to model lateral variations in porosity, permeability, shale volume, and saturation. The present workflow nevertheless demonstrates how heterogeneous geological datasets can be quality-controlled, reconciled, integrated, and visualized to produce a reproducible subsurface characterization.

8. Appendix

The horizon Z values are retained in the horizon-file convention (SSTVD below sea level, positive down).

Hugin Base Distribution:

	line	trace	x	y	z
count	186756	186756	186756	186756	186756
mean	10157.75537	2204.396774	435945.6781	6478433.335	3126.690625
std	112.1325747	138.6338153	1755.698981	1373.020163	165.6042023
min	9963	1959	432192.6686	6475707.782	2741.198975
25%	10061	2086	434531.1175	6477293.071	3001.834229
50%	10154	2205	435957.9169	6478395.428	3142.604248
75%	10253	2322	437422.384	6479559.113	3252.577515
max	10361	2477	439342.9716	6481441.598	3486.77124

Hugin Top Distribution:

	line	trace	x	y	z
count	184066	184066	184066	184066	184066
mean	10159.82279	2203.860164	435958.4391	6478456.787	3058.876513
std	111.2751114	137.6537929	1743.763629	1361.878233	143.1933406
min	9969	1959	432107.8616	6475807.276	2723.337158
25%	10064	2086	434555.857	6477326.135	2946.628235
50%	10157	2204	435976.872	6478421.994	3071.453735
75%	10255	2321	437414.0421	6479574.375	3161.096069
max	10361	2476	439342.9716	6481405.305	3393.910156

Petrophysical survey:

MD	TVD-RKB	TVD SS	x	y	VSH	BVW	SW	PHIF	RHOFL	RHOMA	TEMP	RW	SAND_FLAG	COAL_FLAG	CARB_FLAG	KLOGH	KLOGV
3783	2946.84	2921.84	436366.773	6477641.746	1.7522787 4	0.07536860 63	0.38357007 87	0.196487 7795	0.9	2.66	114.1361047	0.02141759921	0	0	0	0	0
3814	2974.44	2949.44	436371.981	6477628.625	1.5924170 6	0.06374018 11	0.46064173 23	0.138374 7139	0.9	2.66	114.8535635	0.02130487664	0	0	0	0	0
3843	3000.3	2975.3	436376.884	6477616.46	1.0514661 42	0.09277172 441	0.98436535 43	0.094310 55906	0.9	2.66	115.525974	0.02120033228	0	0	0	0	0
3872	3026.23	3001.23	436381.84	6477604.463	1.3387543 31	0.10383366 14	0.93324540 68	0.111263 643	0.9	2.66	116.2001845	0.02109658793	0	0	0	0	0
3900	3051.35	3026.35	436386.629	6477593.071	1.1451685 04	0.10144754 33	0.78570944 88	0.129122 7874	0.9	2.66	116.8533843	0.02099696929	0	0	0	0	0
3929	3077.42	3052.42	436391.605	6477581.377	1.0595249 34	0.08693092 651	0.79393018 37	0.109517 1286	0.9	2.66	117.5311236	0.02089459606	0	0	0	0	0
3957	3102.62	3077.62	436396.481	6477570.19	0.9754314 961	0.12293180 31	1	0.122931 8031	0.9	2.66	118.1862409	0.02079661654	0	0	0	0	0
3987	3129.67	3104.67	436401.71	6477558.334	0.9260173 228	0.11581579 53	1	0.115815 7953	0.9	2.66	118.8896606	0.02069239134	0	0	0	0	0
4015	3155	3130	436406.5	6477547.397	0.2488687 664	0.07960015 223	0.46988346 46	0.169183 5407	0.9	2.66	119.5481197	0.02059582782	1	0	0	13.1080336	8.713630446
4044	3181.26	3156.26	436411.452	6477536.141	0.0487102 3622	0.20383825 98	1	0.203838 2598	0.9	2.66	120.2309157	0.02049658346	1	0	0	1557.898947	1389.314039
4073	3207.37	3182.37	436416.666	6477524.652	0.1024120 735	0.15941030 45	0.81288976 38	0.196556 0446	0.9	2.65	120.9097262	0.02039886693	1	0	0	458.1295643	379.3876522

MD	TVD-RKB	TVD SS	x	y	VSH	BVW	SW	PHIF	RHOFL	RHOMA	TEMP	RW	SAND_FLAG	COAL_FLAG	CARB_FLAG	KLOGH	KLOGV
4102	3233.15	3208.15	436422.331	6477512.648	0.3462958005	0.1190213675	1	0.11902136759021	0.9	2.65	121.5800367	0.02030329475	0	0	0	0.6867763778	0.3803963255
4140	3266.37	3241.37	436430.359	6477496.043	0.1010590551	0.1597578898	1	0.159757 8898	0.9	2.68519685	122.4437047	0.02018152283	0.35433070 87	0	0.6456692913	58.61385433	191.8894236
4170	3292.23	3267.23	436437.01	6477482.377	0.3300496063	0.2440731102	0.9630196 85	0.25 3783 4252	0.9	2.65	123.1160961	0.02008769764	1	0	0	11.65355984	7.692765354
4197	3315.36	3290.36	436443.233	6477469.925	0.9600047244	0.1021341969	0.8425826772	0.12 4524 748	0.9	2.65	123.7175583	0.02000044 5197	0	0	0	0.2621409449	0.1400732283
4226	3340.13	3315.13	436450.115	6477456.504										0	0		

Petrophysical interpretation curves MD with Hugin top and base well picks:

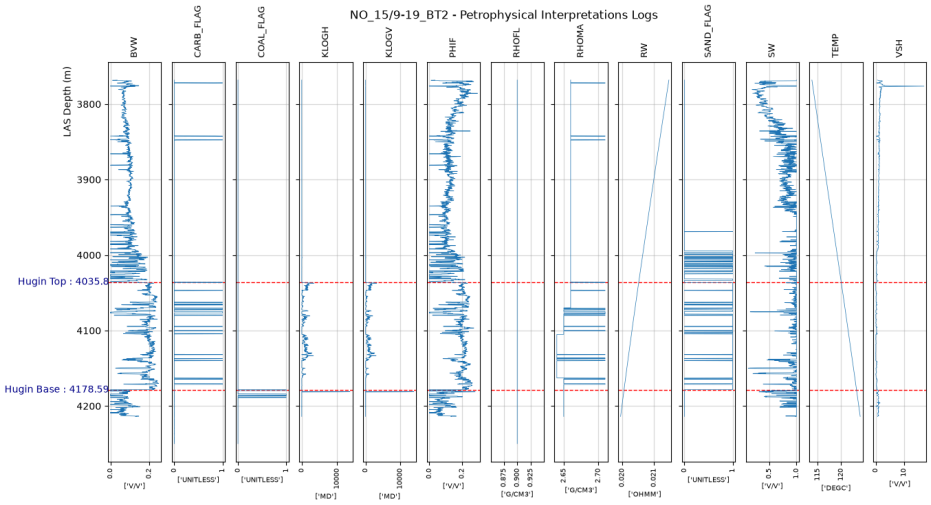


Figure 8.1. 15-9-19 BT2 Well log Petrophysical Interpretations with Hugin Picks interval

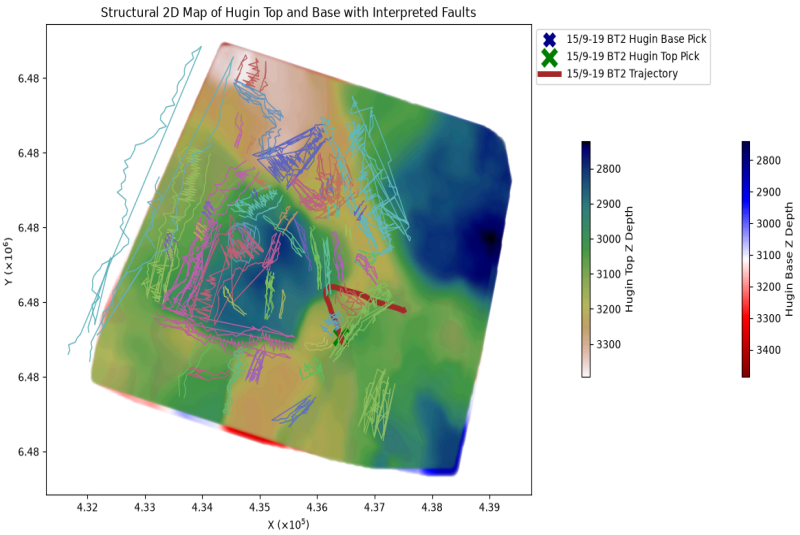


Figure 8.1. 2D 2011 Regional fault-stick interpretation

References

Equinor ASA, Volve Data Village, accessed through Databricks Marketplace / Volve Data Village. Project-specific files used include Hugin top/base horizon interpretations, Volve well picks, 15/9-19 BT2 Standard Survey NPD, Volve Official Faults and 15_9-19_BT2_CPI.las.

Data Access

Original dataset: [Databricks Marketplace - Volve Data Village - Equinor ASA]